A Course-Based Project Report on

SEARCH ENGINE FOR SYNCHROTRON DATA USING THE HADOOP ECOSYSTEM

Submitted to the

Department of CSE-(CyS, DS) and AI&DS

in partial fulfilment of the requirements for the completion of course

Big Data Computing LABORATORY(22PC2DS303)

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science

Submitted by

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An Autonomous Institute, NAAC Accredited with ‘A++’ Grade.

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APRIL’ 2025

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Department of CSE-(CyS, DS) and AI&DS

TAMASOMA JYOTIRGAMAYA

Estd.1995

CERTIFICATE

This is to certify that the project report entitled "**SEARCH ENGINE FOR SYNCHROTRON DATA USING HADOOP ECOSYSTEM**" is a bonafide work done under our supervision and is being submitted by K J Ritiesh (22071A7223), K Dinesh Reddy (22071A7224), K Nipun (22071A7225), M Bhanu Prakash (22071A7234) in partial fulfilment for the award of the degree of Bachelor of Technology in Artificial Intelligence & Data Science, of the VNRVJIET, Hyderabad during the academic year 2024-2025.

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Estd. 1995

DECLARATION

We declare that the course based project work entitled "**SEARCH ENGINE FOR SYNCHROTRON DATA USING HADOOP ECOSYSTEM**" submitted in the Department of **CSE-(CyS, DS) and AI&DS**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology in Artificial Intelligence & Data Science** is a bonafide record of our own work carried out under the supervision of **D. Veera Bhadra Rao, Assistant Professor, Department of CSE (CyS, DS) and AI&DS , VNRVJIET.**. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree from any other institution or university previously..

Place: Hyderabad.

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ACKNOWLEDGEMENT

We express our deep sense of gratitude to our beloved President, **Sri. D. Suresh Babu**, VNR Vignana Jyothi Institute of Engineering & Technology for the valuable guidance and for permitting us to carry out this project.

With immense pleasure, we record our deep sense of gratitude to our beloved Principal, **Dr. C.D Naidu**, for permitting us to carry out this project.

We express our deep sense of gratitude to our beloved Professor **Dr.T.Sunil Kumar**, Professor and Head, Department of CSE-(CyS, DS) and AI&DS , VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad-500090 for the valuable guidance and suggestions, keen interest and through encouragement extended throughout the period of project work.

We take immense pleasure to express our deep sense of gratitude to our beloved Guide, **D. Veera Bhadra Rao**, Assistant Professor in CSE-(CyS, DS) and AI&DS, VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad, for his/her valuable suggestions and rare insights, for constant source of encouragement and inspiration throughout my project work.

We express our thanks to all those who contributed for the successful completion of our project work.

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ABSTRACT

In this project we implemented the design of search engine on top of the Apache Hadoop framework. A synchrotron as an experimental physics facility can provide the opportunity of a multi-disciplinary research and collaboration between scientists in various fields of study such as physics, chemistry, etc. During the construction and operation of such facility valuable data regarding the design of the facility, instruments and conducted experiments are published and stored. It takes researchers a long time going through different results from generalized search engines to find their needed scientific information so that the design of a domain specific search engine can help researchers to find their desired information with greater precision. It also provides the opportunity to use the crawled data to create a knowledgebase and also to generate different datasets required by the researchers. There have been several other vertical search engines that are designed for scientific data search such as medical information. Usage of Hadoop ecosystem provides the necessary features such as scalability, fault tolerance and availability. It also abstracts the complexities of search engine design by using different open-source tools as building blocks, among them Apache Nutch for the crawling block and Apache Solr for indexing and query processing. Our primary results obtained by implementing the proposed method in single node mode, the index of over a hundred thousand pages was created with the average fetch interval of 30 days having 28 segments and approximately 570 MB size. The performance factors such as the usage of available bandwidth and system load were logged using Linux's sysstat package.

CHAPTER-1

INTRODUCTION

Particle accelerators and synchrotrons are critical tools in experimental physics, facilitating research across various disciplines by enabling precise studies at the nanoscopic scale. Synchrotron facilities, in particular, produce synchrotron radiation which is essential for spectroscopy, imaging, and scattering experiments in physics, chemistry, biology, and medicine. These experiments generate vast amounts of data, with estimates reaching up to 50–70 TB per year at facilities like CERN . Handling and analyzing such large datasets require advanced Big Data tools.

The challenge scientists face lies in efficiently searching and accessing relevant datasets and documents related to their research within the synchrotron community. Generalpurpose search engines often fall short due to their inability to refine searches to specific scientific domains. In response to this need, this paper proposes the design of a domainspecific search engine tailored for the synchrotron community. This search engine aims to improve the discoverability and accessibility of scientific data and documents crucial for experimental physicists.

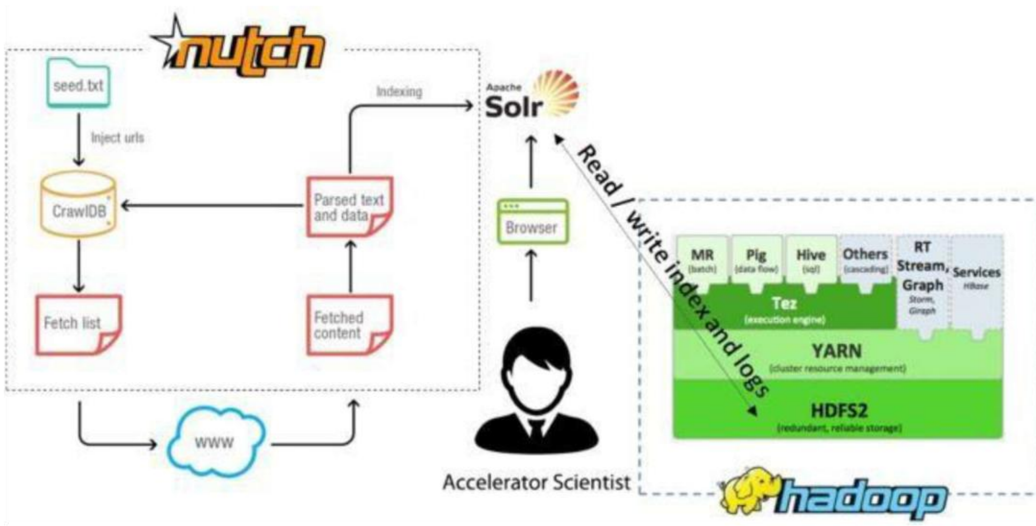
CHAPTER-2

METHODOLOGY

2.1 Overview

The proposed domain-specific search engine leverages the Apache Hadoop ecosystem for scalability and Apache Nutch for web crawling, coupled with Apache Solr for indexing and querying. This section outlines the architecture and key components of the search engine.

2.2 Architecture



2.3 Implementation Steps

Crawling: Apache Nutch is configured to crawl various synchrotron websites, collecting scientific documents and datasets. Apache Nutch which is an extensible and scalable web crawler is used for crawling synchrotron websites to collect the required data. It can be configured to use HDFS providing the scalability for storing huge amount of data using the distributed file system. Nutch has a number of plug-ins that can be used to process a variety of file types such as Plain Text, XML, Open Document, Microsoft Office, PDF, RTF etc. This feature is very useful since it can be used for processing most of the Synchrotron Data formats.

Most researchers design their own implemented web crawlers when it comes to creating a focused crawler. Supervised and semi-supervised machine learning techniques are used to improve relevant document retrieval but there is little literature on the performance of such crawlers for web-scale data. In order to crawl an store specific domains we need to define some URL filters using regular expressions. Figure 3 shows such configuration added to the end of regex-urlfilter text file stored in the Nutch's configuration directory.

```
+^https?://([a-z0-9-]+\.)*albasynchrotron\.es/
+^https?://([a-z0-9-]+\.)*elettra\.eu/
+^https?://([a-z0-9-]+\.)*esrf\.eu/
+^https?://([a-z0-9-]+\.)*als\.lbl\.gov/
+^https?://([a-z0-9-]+\.)*aps\.anl\.gov/
+^https?://([a-z0-9-]+\.)*ansto\.gov\.au/
+^https?://([a-z0-9-]+\.)*helmholtz-berlin\.de/
+^https?://([a-z0-9-]+\.)*lightsource\.ca/
+^https?://([a-z0-9-]+\.)*chess\.cornell\.edu/
+^https?://([a-z0-9-]+\.)*diamond\.ac\.uk/
+^https?://([a-z0-9-]+\.)*lnls\.cnpem\.br/
+^https?://([a-z0-9-]+\.)*maxiv\.lu\.se/
+^https?://([a-z0-9-]+\.)*photon-science\.desy\.de/
+^https?://([a-z0-9-]+\.)*nsrrc\.org\.tw/
+^https?://([a-z0-9-]+\.)*pal\.postech\.ac\.kr/
+^https?://([a-z0-9-]+\.)*bnl\.gov/
+^https?://([a-z0-9-]+\.)*www-ssrl\.slac\.stanford\.edu/
+^https?://([a-z0-9-]+\.)*sesame\.org\.jo/
+^https?://([a-z0-9-]+\.)*kek\.jp/
+^https?://([a-z0-9-]+\.)*slri\.or\.th/
+^https?://([a-z0-9-]+\.)*psi\.ch/
+^https?://([a-z0-9-]+\.)*synchrotron\.uj\.edu\.pl/
```

Indexing: For indexing the fetched documents, Nutch can be configure to send them to Apache Solr which is built on the Apache Lucene information retrieval library. Apache Lucene uses a vector space model method for creating the index and it supports a variety of query types such as fielded term with boosts, wildcards, fuzzy (using Levenshtein Distance), proximity searches and Boolean operators.

Query Parsing: Users interact with the search engine through a custom interface that sends queries to Solar.

User Interface: When data is collected by the Apache Nutch and an Index is generated, users can describe their intent in the form of a Query including several keyword terms.

The user's input query then should be sent to the search engine to retrieve related information. This can be done through creating different user interfaces to capture the query input.

Apache Solar can be accessed through sending a HTTP request containing the user query in the form of field and value. When a query is sent Solar would run the query and send the results in JSON format. This feature enables the UI designer to create a variety of applications for end users of different platforms such as web pages or mobile applications.

2.4 Tools and Technologies

Apache Nutch: Configured to handle various document formats typically found in synchrotron data.

Apache Solar: Provides robust indexing and querying capabilities suitable for scientific data.

Hadoop HDFS: Ensures scalability and fault tolerance for large-scale ata storage.

2.5 System Configuration

Initial implementation was conducted in a single-node setup, indexing approximately 100,000 documents with an average fetch interval of 30 days. Performance metrics such as system load, network traffic, and CPU usage were monitored using sys stats and KSar tools

CHAPTER-3

CODE

1. Apache Nutch Configuration

Assume Apache Nutch is configured to crawl synchrotron websites and store the crawled data in HDFS.

bash Copy code

```bash
# Define seed URLs for crawling in seed.txt
echo "http://example.com" > seed.txt

# Configure Nutch to crawl
bin/nutch inject crawl/crawldb seed.txt
bin/nutch generate crawl/crawldb crawl/segments
bin/nutch fetch crawl/segments
bin/nutch parse crawl/segments
bin/nutch updatedb crawl/crawldb crawl/segments
```

2. Apache Solr Schema Definition

Define a schema.xml file for Solr to specify how data should be indexed.

xml Copy code

```xml
<!-- Example schema.xml -->
<schema name="example" version="1.6">
  <fields>
    <field name="id" type="string" indexed="true" stored="true" required="true"/>
    <field name="title" type="text_general" indexed="true" stored="true"/>
    <field name="content" type="text_general" indexed="true" stored="true"/>
    <field name="url" type="string" indexed="true" stored="true"/>
    <field name="timestamp" type="date" indexed="true" stored="true"/>
  </fields>
  <uniqueKey>id</uniqueKey>
  <defaultSearchField>content</defaultSearchField>
  <solrQueryParser defaultOperator="AND"/>
</schema>
```

3. Indexing with Apache Solr

Assume data from Nutch is indexed into Solr using the defined schema.

bash   Copy code

```bash
# Start Solr (assuming Solr is already installed and configured)
bin/solr start

# Create a new core using the defined schema
bin/solr create -c synchrotron -d path/to/schema.xml

# Index data from Nutch into Solr
bin/nutch solrindex http://localhost:8983/solr/synchrotron crawl/crawldb -linkdb crawl/linkdb
```

4. Querying with Apache Solr

You can query the indexed data in Solr using HTTP requests or through a client library.

bash   Copy code

```bash
# Example query to search for documents containing "synchrotron"
http://localhost:8983/solr/synchrotron/select?q=synchrotron&wt=json&indent=true
```

CHAPTER-4

TEST CASES/OUTPUT

The proposed architecture was implemented in a single node mode, it indexed about a hundred thousand documents with average fetch interval 30 days creating an index with 28 segments and 579 MB size. The performance of the machine was monitored and logged via sys stats package and K Sar. Figure 4 shows the run queue and average load status of the system in 1 h period during the crawl process. The network traffic of the eth0 interface is shown in Fig. 5, the Fig. 6 illustrates the CPU status and the sockets status is shown in Fig. 7.

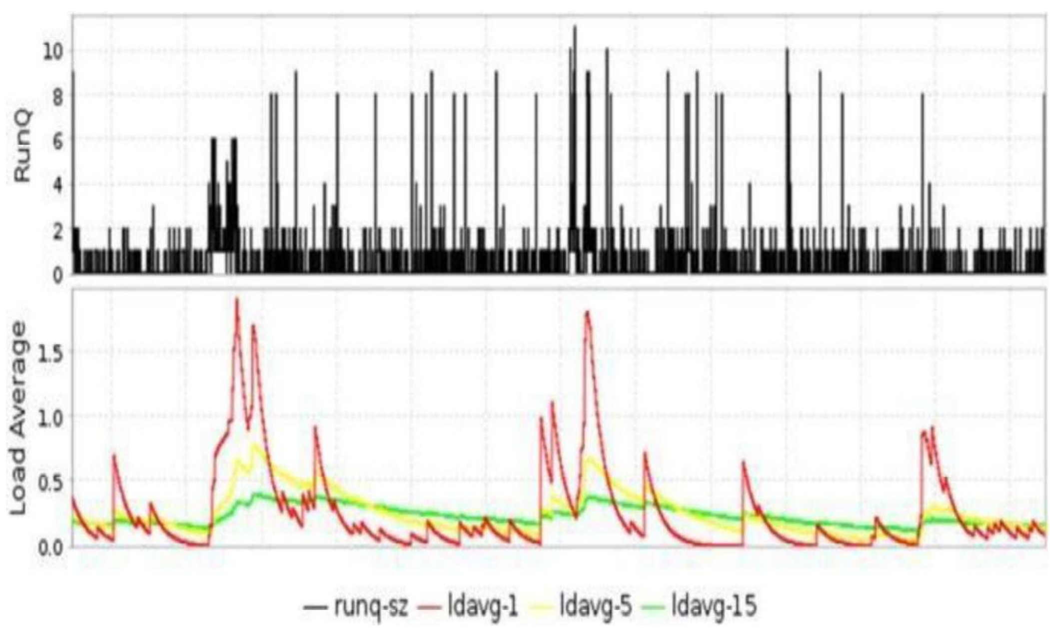


FIG: 4

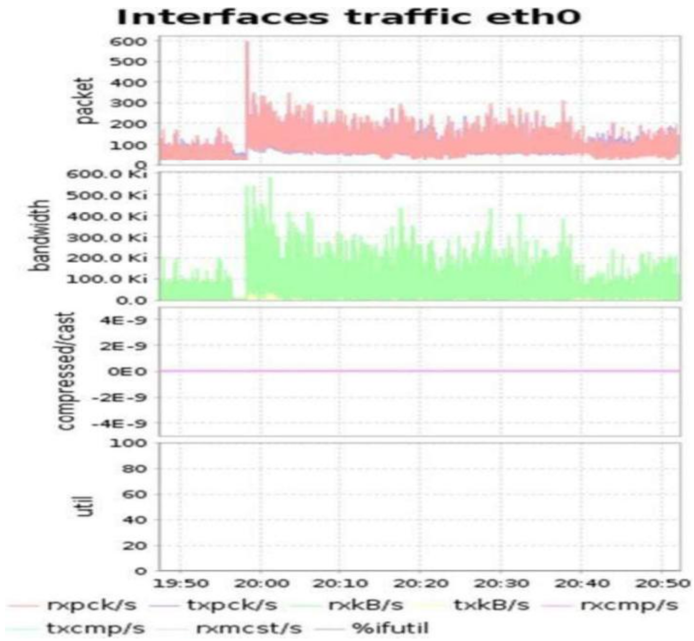
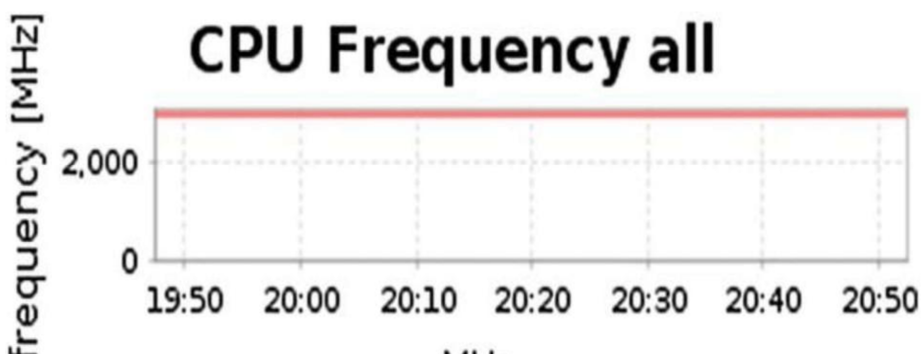

Interfaces traffic eth0
packet
600
500
400
300
200
100
0
bandwidth
600.0 Ki
500.0 Ki
400.0 Ki
300.0 Ki
200.0 Ki
100.0 Ki
0.0
compressed/cast
4E-9
2E-9
0E0
-2E-9
-4E-9
util
100
80
60
40
20
0
19:50
20:00
20:10
20:20
20:30
20:40
20:50
rxpck/s
txpck/s
rxkB/s
txkB/s
rxcmp/s
txcmp/s
rxmcst/s
%ifutil

FIG: 5

FIG: 6


CPU Frequency all
frequency [MHz]
2,000
0
19:50
20:00
20:10
20:20
20:30
20:40
20:50
MHz

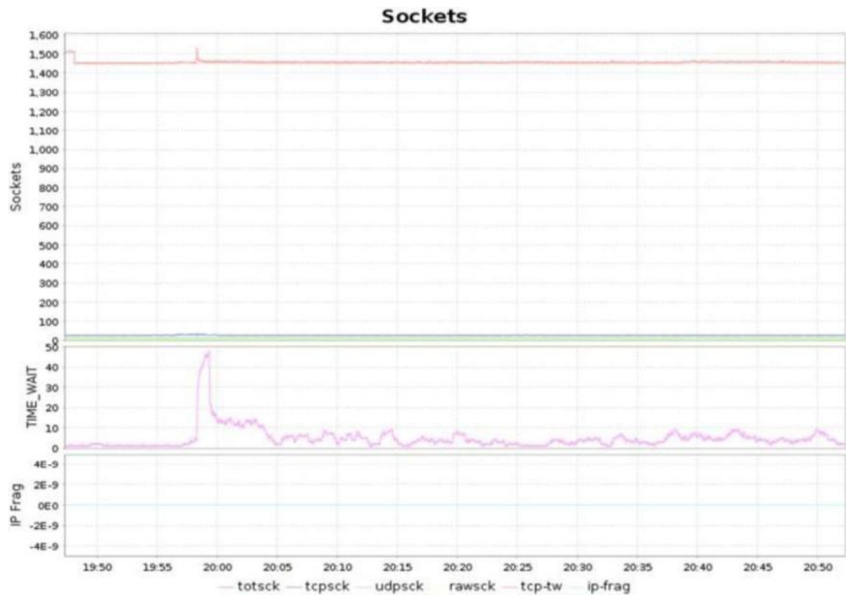

Sockets
Sockets
TIME_WAIT
IP Frag
19:50
19:55
20:00
20:05
20:10
20:15
20:20
20:25
20:30
20:35
20:40
20:45
20:50
totsck
tcpsck
udpsck
rawsck
tcp-tw
ip-frag

FIG: 7

CHAPTER-5

RESULTS

5.1 Performance Metrics

During the crawl process:

System Load: Average load remained below 0.5, peaking during Solr indexing.

Network Traffic: Monitored on eth0 interface, showing consistent bandwidth usage.

CPU Usage: Analyzed across all cores, indicating optimal performance during crawling tasks.

Sockets: Managed effectively to maintain connection integrity during data retrieval.

5.2 Indexing Efficiency

Index Size: Initial indexing created an index with 28 segments totaling 579 MB.

Scalability: Single-node setup demonstrated feasibility; future scalability to a Hadoop cluster is recommended for broader deployment.

CHAPTER-6

Summary, Conclusion, Recommendation

6.1 Summary

This paper proposed a domain-specific search engine tailored for the synchrotron community, leveraging Apache Nutch, Solar, and Hadoop HDFS for data collection, indexing, and querying. Initial implementation demonstrated effective crawling and indexing capabilities, laying the foundation for future enhancements and scalability.

6.2 Conclusion

In conclusion, the domain-specific search engine designed for synchrotron data addresses the specific needs of experimental physicists by improving access to relevant scientific documents and datasets. The use of Apache Hadoop tools provides scalability and robust performance, essential for handling the vast amounts of data generated by synchrotron experiments.

6.3 Recommendation

Future work should focus on:
Cluster Deployment: Transitioning to a multi-node Hadoop cluster for enhanced scalability.

User Interface Enhancement: Developing intuitive interfaces to improve usability.

Data Integration: Incorporating data from additional synchrotron facilities worldwide.

Performance Optimization: Fine-tuning algorithms to optimize crawling and indexing efficiency.

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